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Constrained Image-Based Visual Servoing using Barrier Functions

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***Abstract*— This paper presents a novel constrained image-based visual servoing (IBVS) approach that guarantees target features to remain within the camera field of view (FOV) for the duration of the task. Barrier function (BF) is used to transform the feature dynamics on the image plane into equivalent dynamics with full state constraints. An IBVS controller is designed for the full state constrained system that regulates the original system. A stability analysis is provided to prove the local asymptotic convergence of the state error in the sense of Lyapunov. Simulation and robot implementation results show the efficiency of the proposed method to reach the desired features from any initial configuration while the visibility constraints of the feature points are being realized.**

I. Introduction

Image-based visual servo (IBVS) is a closed-loop control scheme that uses camera image feedback to control the motion of the robot. In the popular eye-in-hand configuration, the camera is mounted near the end-effector of a robot manipulator whose motion can be controlled using the information from the images for tasks such as grasping. In another scenario, the camera mounted on a mobile robot can also use image feedback to regulate its position to a desired goal position or for target tracking applications. It is essential to control the joint motion of the robot such that all the detected feature points remain in the field of view (FOV) for the continuous availability of image feedback. However, maintaining visibility of the feature points is a difficult task to achieve due to multiple factors regarding the geometry of the scene and the camera [1], [2]. The feature points' dynamics are governed by their time-varying distance from the camera, which is an unknown parameter. In traditional IBVS control, approximations such as constant depth is used [3]. Another source of uncertainty is due to inaccurate camera calibration, as intrinsic camera parameters are required for the computation of the image Jacobian. Such uncertainties in the feature dynamics result in undesired transients and deformed camera (robot) trajectories, leading to loss of visibility of the feature points. A standard method to remedy such transients and unwanted trajectories in traditional IBVS is to tune the control gains empirically, which can be laborious and still may not guarantee the loss of features from the FOV. Recently, approaches using switching control which switches between image feedback and predictor methods are proposed in [4], [5]. In [6], a new IBVS controller for leader-follower scenario is developed using the concept of error transformation [7]. In contrast, this paper presents a constrained IBVS method by utilizing advances of barrier function (BF) method [8] to design a camera velocity controller such that the feature points maintain the user prescribed constraints/bounds. The controller stability is shown using Lyapunov analysis.

The unknown parameters in the image Jacobian can be estimated to reduce the uncertainty and improve the IBVS trajectories. In [9]–[13], observer-based methods are presented to estimate the time-varying depth of feature points, which can be used to reduce uncertainty in the image Jacobian. In [14], the authors propose an adaptive tracking controller, which ensures global asymptotic position and orientation tracking using an uncalibrated camera. However, the IBVS trajectories can be constrained to ensure the feature points remain in the FOV without estimating the unknown parameters in the image Jacobian. In [15], an iteratively computed trajectory is tracked to maintain the features in the camera FOV. In [7], [16], [17], the transient response of the error dynamics is constrained through logarithmic transformations to ensure that the feature points remain in the camera FOV. In [1], a task prioritization based control law is developed that maximizes motion perceptibility to keep multiple moving targets in the FOV. In [18], a position-based visual servo (PBVS) and a backward motion switching controller is used to achieve the same objective. A dynamic visibility checking algorithm is proposed in [19] to check if the features stay in FOV before applying control. In [20], an adaptive image-based controller with visibility constraint for leader-follower formation is developed considering mobile robot kinematics. A convex 3D visible set is in [21] introduced for PBVS, and a trajectory in this set is followed to ensure feature visibility. In [22], a constrained optimization method is developed to plan a trajectory which is then tracked using an IBVS controller. The constraints are derived from the workspace objects and FOV requirements for feature points. In [23], a homography-based visual servo controller is developed for nonholonomic mobile robots with FOV constraints.

Barrier Lyapunov function (BLF) is a method used to design controllers for systems with output and state constraints. The constraints are, in general, expressed as the system state or the output bounds. In [24], BLF is defined only to constrain the output or the partial state of the system. The control barrier function (CBF) approach is another method

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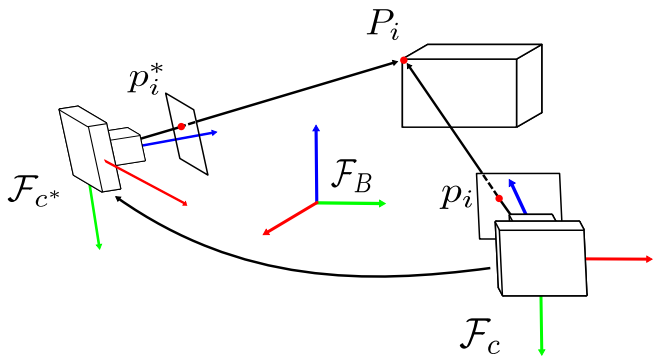

Fig. 1. A moving camera looking at a static scene. $\mathcal{F}_c$ is the current camera frame, $\mathcal{F}_{c^*}$ is the desired coordinate frame, P_i is a static point, p_i is the current projection of that point, p_i^* is the desired projection of that point.

to control nonlinear systems. CBF uses a Lyapunov-like function to certify a control policy that achieves forward invariance of a safe set [25]–[28] by evaluating if solutions of the dynamical system remain inside a compact set for all time. The CBF approach uses numerical optimization to synthesize the controller. In contrast, this paper utilizes barrier transformation to constrain the full state of the IBVS system, i.e., all the feature point states used in the IBVS task.

This paper focuses on the development of a constrained IBVS controller that guarantees the desired features never leave the camera FOV. A BF is used to transform the feature dynamics with full-state visibility constraints to an equivalent system dynamics with no constraints. An IBVS controller is developed for the transformed dynamics that, through the stability analysis, is proven to be locally asymptotically stable. The proposed method is evaluated through numerical simulations, and its performance is demonstrated on a Baxter robot to illustrate the method's ability to adapt to real-world applications.

Throughout this paper, for ease of notation, the time dependency argument of the variables is dropped, e.g., $p_i(t) = p_i$, unless necessary for clarity.

II. Perspective Camera Motion Model and Unconstrained IBVS

Consider a moving camera attached to the robot arm in eye-in-hand configuration, viewing objects in a static scene. Let $\mathcal{F}_c$ be a right handed orthogonal coordinate system attached to the camera such that the Z axis of $\mathcal{F}_c$ is collinear with the optical axis of the camera and the origin of $\mathcal{F}_c$ is at the principal point of the camera. Let $P_i \in \mathbb{R}^3$ be the Euclidean coordinates of the static point in $\mathcal{F}_c$ and $\bar{P}_i \in \mathbb{R}^3$ be the normalized Euclidean coordinates (see Fig. 1).

$$P_i = [X_i, Y_i, Z_i]^T, \quad \bar{P}_i = \left[\frac{X_i}{Z_i}, \frac{Y_i}{Z_i}, 1\right]^T \tag{1}$$

An image plane feature point can be constructed from the normalized Euclidean coordinates such that $p_i = [\frac{X_i}{Z_i}, \frac{Y_i}{Z_i}] \in \mathcal{S}_2 \subset \mathbb{R}^2$. The objective of IBVS is to regulate image plane feature error to zero. Let the state vector for the current features and goal feature locations in the image plane be denoted by $s(t) = \begin{bmatrix} p_1(t)^T & \cdots & p_m(t)^T \end{bmatrix}^T \in \mathcal{S}_{2m} \subset \mathbb{R}^{2m}$ and $s^* = \begin{bmatrix} p_1^{*T} & \cdots & p_m^{*T} \end{bmatrix}^T$, where $m > 3$ is the number of feature points. The dynamics of the multiple feature points s can be expressed as

$$\dot{s} = L_s(s, Z)\,\xi \tag{2}$$

where $L_s \in \mathbb{R}^{2m\times 6}$ is termed as the interaction matrix, $\xi = \begin{bmatrix} v^T & \omega^T \end{bmatrix}^T$ is a vector of the linear and angular camera velocities expressed in the camera reference frame, i.e., $\mathcal{F}_c$, such that $v \in \mathbb{R}^3$ and $\omega \in \mathbb{R}^3$, and $Z = [Z_1 \ \cdots \ Z_m]^T$ are the time-varying depth of the feature points in the camera reference frame. The inertial reference frame, $\mathcal{F}_B$, considered in this paper is the base frame of the robot. The camera control velocities ξ are computed relative to the base frame by suitable rotation matrix for experimental purposes. In the feature dynamics (2), $L_s = \begin{bmatrix} L_{s_1}^T & \cdots & L_{s_m}^T \end{bmatrix}^T$, where

$$L_{s_i} = \begin{bmatrix} \frac{-1}{Z_i} & 0 & \frac{p_{i,1}}{Z_i} & p_{i,1}p_{i,2} & -(1+p_{i,1}^2) & p_{i,2} \\ 0 & \frac{-1}{Z_i} & \frac{p_{i,2}}{Z_i} & 1+p_{i,2}^2 & -p_{i,1}p_{i,2} & -p_{i,1} \end{bmatrix}, \tag{3}$$

with $p_{i,1}(t)$ and $p_{i,2}(t)$ denote the first and second element of feature point $p_i(t)$, $\forall i = 1, \cdots, m$. The corresponding error vector between the current and the goal configuration is the defined as $e(t) = s(t) - s^*$. The derivative of the error vector can be expressed in terms of camera velocity as

$$\dot{e} = L_s \xi. \tag{4}$$

Assuming that 3 or more feature points are available in the FOV, the unconstrained IBVS control law is given by

$$\xi = -\gamma \widehat{L}_s^{\dagger} e \tag{5}$$

where $\gamma \in \mathbb{R}^+$ is a constant and $\widehat{L}_s$ is an approximation of L_s as the true depth Z_i, $\forall i = 1, \cdots, m$ is unknown. The right pseudo-inverse of $\widehat{L}_s$ is given by $\widehat{L}_s^{\dagger} = (\widehat{L}_s^T \widehat{L}_s)^{-1} \widehat{L}_s^T$. The control law ensures local asymptotic stability when $\widehat{L}_s^{\dagger} L_s > 0$, i.e., 3 or more feature points are available in the FOV for all time. If the feature points leave the FOV, the controller will fail.

A Barrier function based transformation is used in the following section to design the camera velocity controller such that the feature vectors s are regulated to the desired constant feature vectors s^* with an additional objective that they remain inside the camera FOV by satisfying a user prescribed state constraints for all time.

III. Barrier Functions

Consider a locally Lipschitz continuous nonlinear dynamical system of the following form

$$\dot{p}_i(t) = L_{s_i} \xi. \tag{6}$$

The following logarithmic barrier function candidate $B(p_{i,j}; a_{i,j}, A_{i,j}) : \mathbb{R} \to \mathbb{R}$, is considered in this paper:

$$B(p_{i,j}; a_{i,j}, A_{i,j}) \triangleq \log\left(\frac{A_{i,j}}{a_{i,j}} \frac{a_{i,j} - p_{i,j}}{A_{i,j} - p_{i,j}}\right), \tag{7}$$

$\forall p_{i,j} \in (a_{i,j}, A_{i,j})$, where $a_{i,j} \in \mathbb{R}$ and $A_{i,j} \in \mathbb{R}$ are lower and upper bounds of each feature point, $\log(\cdot)$ is a

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natural logarithm, and $j = 1, 2$. From (7), it can be seen that $\lim_{p_{i,j} \to a_{i,j}, A_{i,j}} B(p_{i,j}; a_{i,j}, A_{i,j}) = \pm\infty$. Using (7), the system in (6) can be transformed into a constrained state $x_i = [x_{i,1}, x_{i,2}]^T \in \mathcal{S}_2 \subset \mathbb{R}^2$ as follows

$$x_{i,j} = B(p_{i,j}; a_{i,j}, A_{i,j}), \tag{8}$$

$$p_{i,j} = B^{-1}(x_{i,j}; a_{i,j}, A_{i,j}), \tag{9}$$

$\forall i = 1, \cdots, m$ and $\forall j = 1, 2$, where

$$B^{-1}(x_{i,j}; a_{i,j}, A_{i,j}) = \frac{a_{i,j} A_{i,j} \left(-e^{\frac{-x_{i,j}}{2}} + e^{\frac{x_{i,j}}{2}}\right)}{-A_{i,j} e^{\frac{-x_{i,j}}{2}} + a_{i,j} e^{\frac{x_{i,j}}{2}}}. \tag{10}$$

Note that due to the monotonic characteristic of the natural logarithm the inverse function (9) exists within the range of its definition.

Using the chain rule of differentiation, i.e., $\frac{dp_{i,j}}{dt} = \frac{\partial p_{i,j}}{\partial x_{i,j}} \frac{dx_{i,j}}{dt}$, where

$$\frac{\partial p_{i,j}}{\partial x_{i,j}} = \frac{A_{i,j} a_{i,j}^2 - a_{i,j} A_{i,j}^2}{a_{i,j}^2 e^{x_{i,j}} - 2 a_{i,j} A_{i,j} + A_{i,j}^2 e^{-x_{i,j}}} \tag{11}$$

after some algebraic manipulations the new dynamics in $x_{i,j}$ yields

$$\dot{x}_i = K_i L_{s_i} \xi, \tag{12}$$

where $K_i \in \mathbb{R}^{2\times 2}$ is given by

$$K_i = \operatorname{diag}(k_{i,1}, k_{i,2}) \tag{13}$$

for $k_{i,j} = \left(\frac{\partial p_{i,j}}{\partial x_{i,j}}\right)^{-1}$, and $L_{s_i}\left(\left[B^{-1}(x_{i,1}), B^{-1}(x_{i,2})\right], Z_i\right)$ is given by (3).

Let $X = \left[x_1^T, \cdots x_m^T\right]^T \in \mathbb{R}^{2m}$ be a state vector constructed using a collection of constrained feature point state. By collecting the individual constrained state dynamics in (12), the dynamics of the constrained feature point state X is obtained

$$\dot{X} = K L_s \xi, \tag{14}$$

where $K = \operatorname{blkdiag}(K_1, \cdots, K_m) \in \mathbb{R}^{2m\times 2m}$ with K_i given in (13).

Remark 1: The upper and lower bounds for feature points can be selected using the perspective projection formulae, i.e., $a_{i,1} = \frac{\underline{u} - c_x}{f_x}$, $A_{i,1} = \frac{\bar{u} - c_x}{f_x}$, $a_{i,2} = \frac{\underline{v} - c_y}{f_y}$, $A_{i,2} = \frac{\bar{v} - c_y}{f_y}$. $(\underline{u}, \underline{v})$ and $(\bar{u}, \bar{v})$ are lower and upper bounds on pixel coordinates, which can be selected based on the image size. (c_x, c_y) is the center of the image, and f_x, f_y are the camera focal lengths.

IV. Constrained IBVS Controller

A. Control Development

A camera velocity controller is developed in this section to regulate the state $X(t)$ to a constant value X^*. To achieve the control objective, a controller is developed using the transformed state space dynamics (14).

For the system in (14), consider the multiple feature error vector $e_x(t) \in \mathbb{R}^{2m}$ defined by

$$e_x(t) = X(t) - X^*, \tag{15}$$

where the vector X^* contains the desired values of the features in the constrained space. Taking the derivative of (15) with respect to time and substituting for (14), the following error dynamics is obtained:

$$\dot{e}_x(t) = K L_s \xi. \tag{16}$$

The control input is designed to compute camera velocities such that the feature points are aligned with the desired feature points in the constrained state space. The designed velocity control input is given by

$$\xi = -\gamma \widehat{L}_s^{\dagger} K^T e_x. \tag{17}$$

Substituting the controller (17) in (16) yields the following closed-loop dynamics

$$\dot{e}_x(t) = -\gamma K L_s \widehat{L}_s^{\dagger} K^T e_x. \tag{18}$$

B. Stability Analysis

The stability result of the closed-loop error dynamics (18) derived using Lyapunov stability analysis is stated in the following Theorem.

Theorem 1: Given the velocity controller in (17) the closed-loop error dynamics in (18) is locally asymptotically stable.

Proof: Consider the auxiliary error system

$$e' = \widehat{L}_s^{\dagger} K^T e_x. \tag{19}$$

Taking the time derivative of the auxiliary error system and using (16) and (17) yields

$$\begin{aligned} \dot{e}' &= \widehat{L}_s^{\dagger} K^T \dot{e}_x + \dot{\widehat{L}}_s^{\dagger} K^T e_x + \widehat{L}_s^{\dagger} \frac{dK^T}{dX} e_x \circ \dot{X} \\ &= -\gamma \widehat{L}_s^{\dagger} K^T K L_s e' + \dot{\widehat{L}}_s^{\dagger} K^T e_x - \gamma \widehat{L}_s^{\dagger} \frac{dK^T}{dX} e_x \circ K L_s e' \\ &= -\gamma \left(\widehat{L}_s^{\dagger} K^T K L_s + \mathcal{G}(e_x)\right) e' \end{aligned} \tag{20}$$

where $\circ$ denotes the Hadamard product and $\mathcal{G}(e_x) \in \mathbb{R}^{6\times 6}$ is a matrix, which is equal to $\mathbf{0}$ at $e_x = 0$. Using Lyapunov analysis of Theorem 4.7 of [29], the error system in (20) is locally asymptotically stable with respect to the equilibrium point $e_x = e' = 0$ if the linearized system matrix $-\widehat{L}_s^{\dagger} K^T K L_s\big|_{s=s^*}$ is Hurwitz. Using the fact that $L_s \widehat{L}_s^{\dagger} \geq 0$ in a small region around the origin, $\operatorname{rank}\{L_s \widehat{L}_s^{\dagger}\} = 6$, $\operatorname{rank}\{K\} = 2m$, and $K = K^T$, it can be concluded that $K^T L_s \widehat{L}_s^{\dagger} K\big|_{s=s^*}$ has 6 positive eigenvalues (cf. [30]) and the rest of $2m - 6$ eigenvalues are zeros. Using Theorem 1.3.2 in [30], the 6×6 matrix $-\widehat{L}_s^{\dagger} K^T K L_s\big|_{s=s^*}$ has the same eigenvalues as $-K^T L_s \widehat{L}_s^{\dagger} K\big|_{s=s^*}$ without the $2m - 6$ zeros. Using the previous two arguments it can be concluded that all eigenvalues of $-\widehat{L}_s^{\dagger} K^T K L_s\big|_{s=s^*}$ are negative; thus it is Hurwitz. As a result, $\exists \delta > 0$ such that if $\|e'(0)\| < \delta$ then $\lim_{t\to\infty} \|e'(t)\| = 0$ and for a sufficiently small δ, $e' = \widehat{L}_s^{\dagger} K^T e_x \neq 0$ if $e_x \neq 0$. The configuration $K^T e_x \in \operatorname{Ker}\left\{\widehat{L}_s^{\dagger}\right\}$ corresponds to local minima and such a configuration does not exist in the local region of attraction. ■

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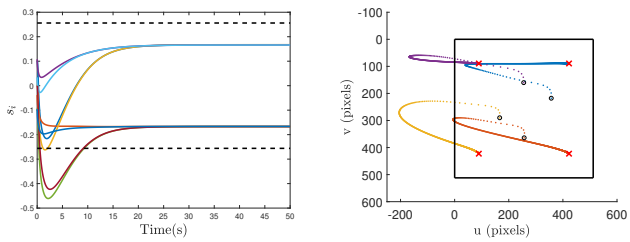
Remark 2: In general, it is difficult to estimate this local region of convergence. The matrix $\widehat{L}_s^{\dagger} K^T K L_s$ can be made full rank by selecting the appropriate feature points (greater than or equal to 3), and by making reasonable approximations in $\widehat{L}_s$.

Remark 3: Using similar arguments of the result of Lemma 3 of [31], the controller designed in (17) regulates the feature point state $s(t)$ of the dynamics (2) to the desired feature points s^* while maintaining the FOV constraints given that the $s(t_0)$ are in the FOV.



(a) (b)

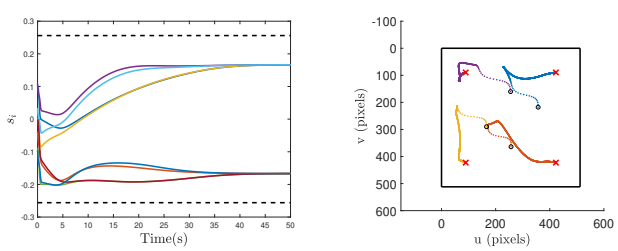
Fig. 2. Simulation results for the IBVS controller without barrier transformation. (a) Image plane feature values along with dashed lines showing the barrier limits. (b) Pixel trajectories with the box indicating image size.

V. Simulation and Experimental Validation

A. Simulation

A simulation is carried out to verify the efficiency of the developed constrained IBVS controller. The simulation is performed in MATLAB 2018a using the Visual Servoing Toolbox in [32] for a total of 50 s with standard four $(m = 4)$ co-planar feature points for the IBVS task. The feature point locations with respect to the inertial frame $\mathcal{F}_B$ whose origin is at $[0, 0, 0]^T$ are $P_1 = [-0.25, 0, 0.25]^T$, $P_2 = [0.25, 0, 0.25]^T$, $P_3 = [-0.25, 0, -0.25]^T$ and $P_4 = [0.25, 0, -0.25]^T$. Note that the known 3D locations of the feature points are not used in the IBVS task. A desired pose for the camera is randomly selected in the inertial frame $\mathcal{F}_B$ such that the feature points are visible in the image when the camera is at the desired pose. The positions of the desired image-plane feature coordinates based on the goal pose are

$$s^* = [0.1667, -0.1667, 0.1667, 0.1667, \\ -0.1667, 0.1667, -0.1667, -0.1667]^T \tag{21}$$



(a) (b)

Fig. 3. Simulation results for the IBVS controller with barrier transformation. (a) Image plane feature values along with dashed lines showing the barrier limits. (b) Pixel trajectories with the box indicating the image size.

The initial image plane feature points are given by

$$s(t_0) = [0.1010, -0.0383, 0.0009, 0.1080 \\ -0.0896, 0.0340, -0.0008, -0.0958]^T \tag{22}$$

The image resolution is assumed to be 512×512. A focal length of 1000 pixels in the X and Y direction is used and the center pixel is set to $(256, 256)$. The sampling time for the simulation is chosen to be 0.03s corresponding to 30Hz frequency which is a reasonable assumption for the current camera hardware. Based on the camera calibration matrix C and the resolution of the image, the bounds for all the image-plane feature points are chosen to be $a_{i,j} = -0.256, A_{i,j} = 0.256 \quad \forall i = 1, \cdots, m$ and $j = 1, 2$. The depth in the approximated interaction matrix $\widehat{L}_s$ is set to 0.9m. Figs 2(a) and 3(a) show the image plane feature trajectories with traditional and constrained IBVS controllers respectively. The dashed black lines show the bounds on the image plane features derived from the image size and the calibration matrix. Fig. 2(a) shows that with the non-constrained IBVS controller, the image plane features cross the bounds and leave the FOV. At this point, a real servoing task is expected to fail. However, for the demonstration, the task is continued until the feature points converge the desired feature points. On the other hand, as it is shown in Fig. 4(a), the barrier constrained IBVS controller deforms the camera trajectory in such a way that the image plane features never cross the dashed black lines. As a result, the features always remain in the FOV. A 2D image pixel depiction of the feature trajectories is shown in Figs. 2(b) and 3(b). The box drawn with black lines represents the size of the image and features inside the box are in the FOV. The image feature pixel trajectories leave the box and hence the FOV as seen from Figure 2(b). However, with the IBVS controller in the constrained space, shown in Fig. 3(b), the image feature pixel trajectories always stay inside the box and hence remain in the FOV.

B. Robot Implementation

To demonstrate the utility of the proposed method in practice, two sets of robot experiments are carried out. Both experiments use a wrist-mounted camera on the right arm of a Baxter research robot to observe a planar target at a rate of 30 fps with resolution 640×400 as depicted in Fig. 6. In the first robot experiment, the target consists of four white circles, which are used against a black background for easy thresholding based image segmentation. The white circles are segmented from the background and the centroids are used as feature points for IBVS. The image processing and controller implementation is carried out in real-time using a desktop with Intel Core2Duo CPU with a clock speed of 2.26 GHz and 4 GB RAM running Ubuntu 14.04. The algorithm is implemented in MATLAB 2018a and runs at 30 Hz matching the camera frequency. The camera calibration matrix for Baxter's right-hand camera obtained through the Baxter API

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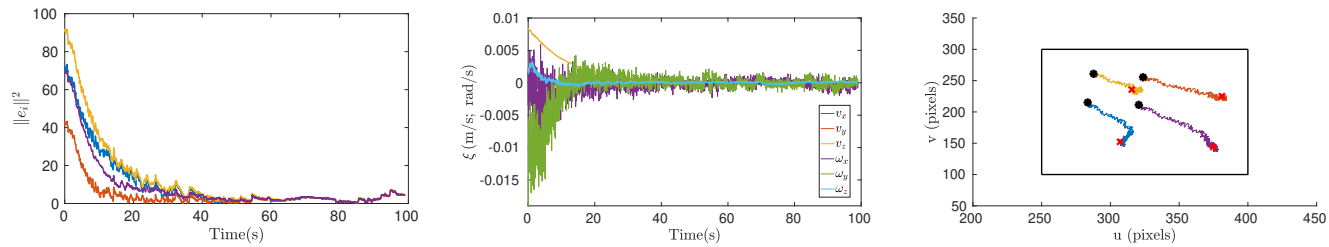
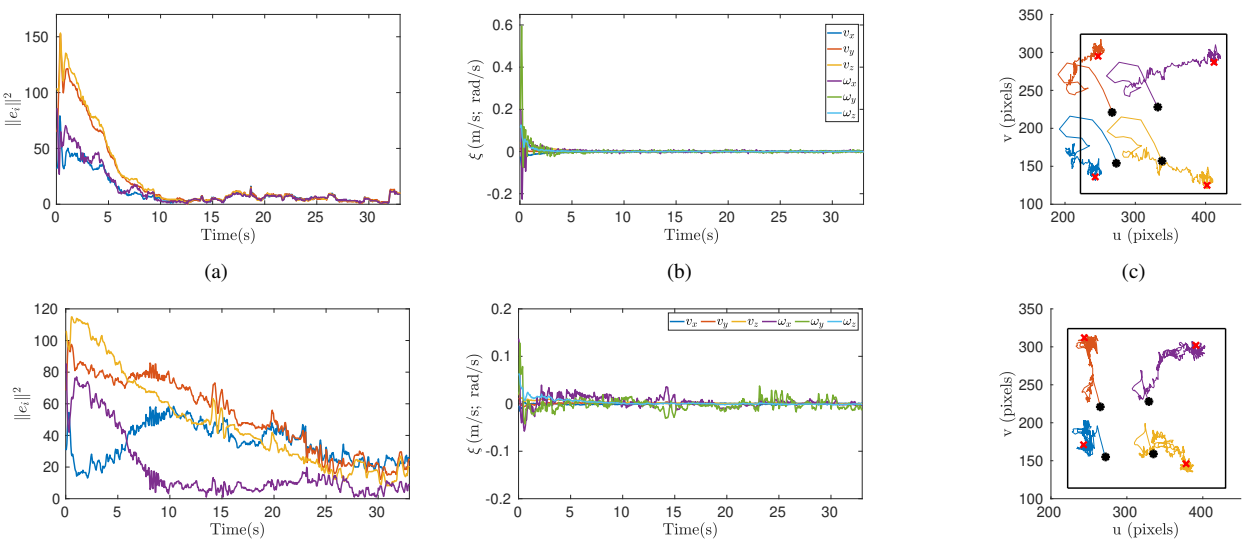


(a) (b) (c)

Fig. 4. Experimental results for the IBVS controller with barrier transformation applied on four white circles laid on a black surface, where the center of each circle denotes the feature image coordinates. (a) Feature error norm. (b) Camera velocities (c) Pixel trajectories with the box indicating the image size.



(a) (b) (c)

(d) (e) (f)

Fig. 5. Experimental results for the IBVS controller without (top row) and with (bottom row) barrier transformation applied on Aruco's marker. (a,d) Feature error norm for four features. (b,e) Camera velocities (c,f) Pixel trajectories with an arbitrarily selected bounding box.

and Robot Operating System (ROS) is given by

$$C = \begin{bmatrix} 407.1 & 0 & 323.4 \\ 0 & 407.1 & 205.6 \\ 0 & 0 & 1 \end{bmatrix} \tag{23}$$

The results of the robot experiment are summarized in Fig. 4(a-c). The error norm of the individual feature points are shown in Fig. 4(a). The error norms of the individual error decrease asymptotically and converge at 65s. The corresponding linear and angular velocities are displayed in Fig. 4(b). The image pixel trajectories are shown in Fig. 4(c). The pixel trajectories remain within the chosen lower and upper bound for the u and v pixel coordinates. The upper and lower bound for all the u coordinates is chosen as 400 and 250, respectively. The upper and lower bounds for all the v pixels are chosen to be 300 and 100, respectively. To implement the control law for image plane coordinates, the bounds in pixel units are converted to image plane bounds using the calibration matrix.

In the second robot experiment, the target consists of a printed Aruco marker [33] placed on the table in front of the Baxter research robot. The experimental setup can be seen in Fig. 6. The corners of the Aruco marker are used as the feature points, and the pixel coordinates of these corners are obtained at every iteration of the algorithm using the Aruco module in OpenCV. The barrier is chosen to have an upper and lower bound on the u coordinates of 430 and 222, and an upper and lower bound on the v coordinates of 324 and 114. The initial robot arm configuration is placed so that the feature points are very close to the lower bound on the v coordinates. Two experiments are run using this setup: one without the barrier and one with the barrier. In both experiments, the desired feature points are chosen as

$$s^* = [-0.2367, -0.2065, 0.1489, -0.2335, \\ 0.1760, 0.1644, -0.2268, 0.1840]^T \tag{24}$$

The control gain is set to $\gamma = 0.2$, and was tuned experimentally. The results of the experiment using the Aruco markers can be seen in Fig. 5, where the top row (a-c) are the results from the experiment without the barrier, and the bottom row (d-f) are the results of the experiment with the barrier. In Fig. 5(a) and 5(d) the error between the desired feature point and the current feature point can be seen for each feature point. The error norms of the feature point error decrease

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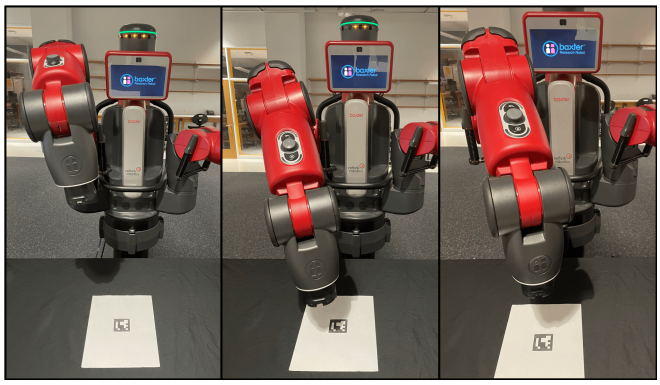

Fig. 6. The experimental setup showing Baxter's end-effector with an eye-in-hand configuration observing the Aruco's marker while moving from an initial camera pose (left-most picture) to the desired camera pose (right-most picture).

to zero asymptotically. In Fig. 5(b) and 5(e) the applied camera control velocities are shown. During the experiment, all control velocities are calculated in the camera frame, and then are converted to the joint space by using the inverse Jacobian of the Baxter robot right arm. The corresponding joint velocities are then applied to each respective arm joint. In Fig. 5(c) and 5(f) the trajectories of each feature point in the image plane are displayed, with the initial pixel values of the feature point shown as black dots and the desired pixel values shown as red $\times$'s. In Fig. 5(c), the trajectory of the purple and yellow feature points can clearly be seen leaving the barrier region, as there is no barrier function applied to this controller. In Fig. 5(f), all feature points remain inside of the barrier region throughout the duration of the experiment.

VI. Conclusion

A constrained IBVS controller is developed in this paper which keeps the feature points within the camera FOV. A logarithmic barrier function transformation is used to derive a new image feature state dynamics. A controller is developed using the constrained state space dynamics for which a local asymptotic stability is shown using Lyapunov stability analysis. The results of the new controller are compared to the unconstrained IBVS controller in simulation. The results demonstrated the ability of the developed controller to maintain all the image plane feature points in the FOV in contrast to the unconstrained IBVS controller. A robot experiment is carried out to demonstrate the results of the constrained controller using the wrist camera of the Baxter robot.

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